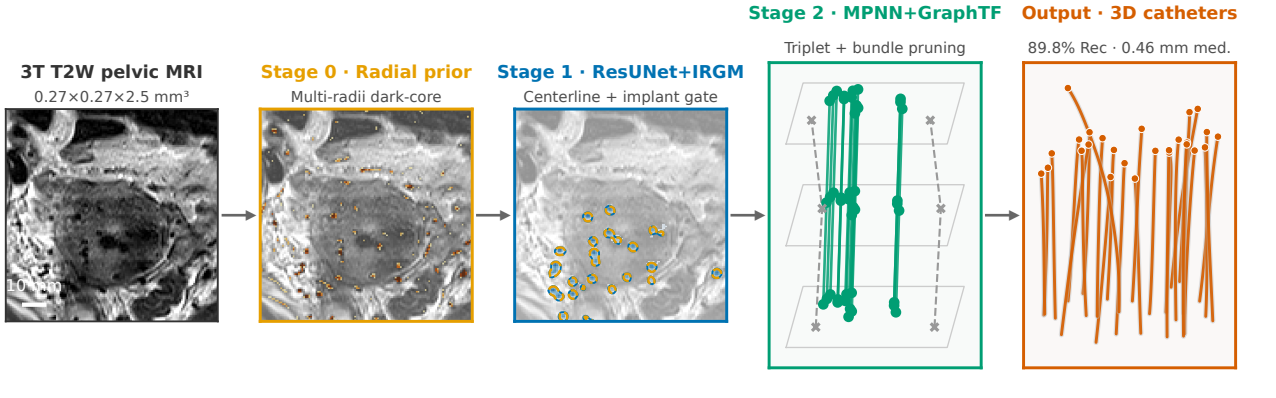

3T T2W pelvic MRI
0.27×0.27×2.5 mm³
10 mm
Stage 0 · Radial prior
Multi-radii dark-core
Stage 1 · ResUNet+IRGM
Centerline + implant gate
Stage 2 · MPNN+GraphTF
Triplet + bundle pruning
Output · 3D catheters
89.8% Rec · 0.46 mm med.
Stage 0: Symmetry (60 ms)
Stage 1: ResUNet+IRGM (455 ms)
Stage 2A: MPNN (65 ms)
Stage 2B: GraphTF (30 ms)
CPU (0.61 s)
0.61 s total (8-core CPU)
GPU (0.35 s)
0.35 s total (RTX 4090 GPU)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
End-to-end inference time per patient volume (s)